# Lesson 2.1 Guided Notes

## Session 1 · Statistical Questions

I can tell whether a question is statistical by deciding if the answers will vary. · 6.SP.A.1

| Word / Spanish | Meaning — complete the blank | Example |
|---|---|---|
| statistical question / pregunta estadística | A question whose answers are expected to _____. | How many pets does each student have? |
| variability / variabilidad | How _____ the answers in a data set are. | 0, 2, 5, 1 pets |
| data / datos | Facts or _____ you collect to answer a question. | 6, 8, 9 hours of sleep |

### Key idea

A statistical question expects answers that _____, so you collect data to answer it.

A question with one _____ answer is not statistical, even if the answer is a _____.

### Worked example

Is “How many minutes does each student ride the bus?” a statistical question?

Who answers? _____ student.

Will the answers differ? Yes: 5, 12, 25 minutes, so the answers _____.

Answer: The question is _____.

### You try

1. Write S (statistical) or N (not statistical).
a. How many days are in May? _____
b. How tall is each player? _____

_____

2. “How tall is our classroom door?” is not statistical. Rewrite it so the answers vary.

_____

Sum it up: A question is statistical when ___ because ___.

_____

# Session 2 · Dot Plots and Shape

I can build a dot plot and describe the shape of the data it shows. · 6.SP.B.4

| Word / Spanish | Meaning — complete the blank | Example |
|---|---|---|
| dot plot / diagrama de puntos | A number line with one _____ for each data value. | 3 dots above 2 = three 2s |
| peak / pico | The _____ stack; it shows the most common value. | tallest stack above 5 |
| symmetric / simétrico | The left and right sides of the plot _____. | stacks 1, 3, 5, 3, 1 |
| skewed / sesgado | The data has a long _____ on one side. | tail to the right: skewed right |

## Key idea

Repeated values _____ up, so the total number of dots equals the number of _____.

A plot is named for its _____: a tail to the left means skewed left.

## Worked example

Six friends ate 4, 5, 5, 6, 5, 7 grapes. Make a dot plot.

Draw a line from 3 to 8 and label each whole number.

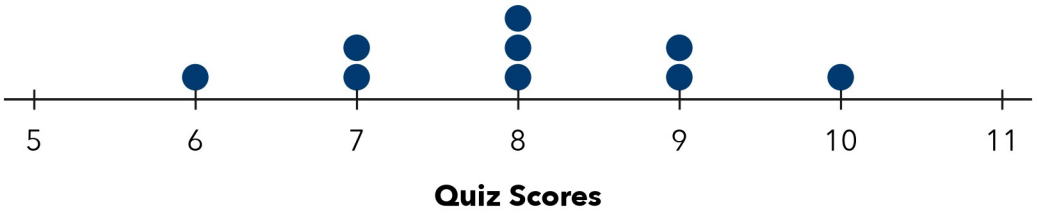
Stack one dot per friend: _____ dots above 5.

The peak is at _____; there are _____ dots in all.

Answer: Most common value: _____ grapes

## You try

1. The plot shows quiz scores. Name the most common value and the shape.




**Quiz Scores**

_____

2. A dot plot has stacks of 2, 4, 1 and 3 dots.
a. Number of values: _____
b. Tallest stack: _____

_____

Sum it up: A dot plot shows ___, so I can see ___.

_____

# Session 1 · Model, reason, and explain

Statistical Questions · 6.SP.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Sort these questions: A. How many minutes did one runner take on each of six practice days? B. What is the runner's recorded time on Monday? Identify the variable and explain which question anticipates variability.

| Population / occasions | Measured variable | Unit | Anticipated variation |
|---|---|---|---|
| | | | |

_____

_____

_____

## Extend your reasoning

Write a statistical question about classroom objects, and name the population and unit.

_____

_____

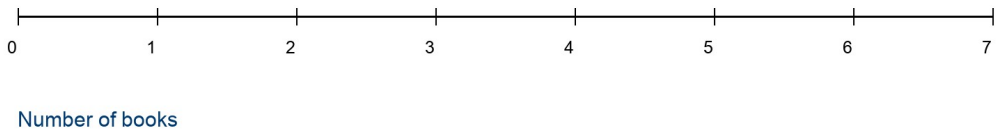
Self-check: Did my model, calculation, explanation, and units agree?

# Session 2 · Model, reason, and explain

Dot Plots and Shape · 6.SP.B.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Six observations are 2, 3, 3, 3, 5, 6 books. Complete the dot plot. Identify the peak and a gap. Explain how you know all observations are represented.




_____

_____

_____

## Extend your reasoning

Can adding one observation create a second value tied for the greatest frequency? Explain.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?